Alexander Y. Ku, Curriculum Vitae

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Education

2028	PhD, Princeton University (in Psychology & Neuroscience) *Advisors:* Thomas L. Griffiths, Jonathan D. Cohen
2025	MA, Princeton University (in Psychology)
2018	MS, UC Berkeley (in Electrical Engineering & Computer Science)
2017	BA, UC Berkeley (in Computer Science)

Experience

2018–	Research Scientist, Google DeepMind
2018	Research Intern, Google Brain
2017	Research Intern, Google Brain

Publications

Journal articles

[1] Alexander Ku, Declan Campbell, Xuechunzi Bai, Jiayi Geng, Ryan Liu, Raja Marjieh, R. Thomas McCoy, Andrew Nam, Ilia Sucholutsky, Liyi Zhang, Jian-Qiao Zhu, and Thomas Griffiths. “Levels of analysis for large language models”. In: *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 384.2320 (May 2026), p. 20250012.

[2] Jiahui Yu, Yuanzhong Xu, Jing Yu Koh, Thang Luong, Gunjan Baid, Zirui Wang, Vijay Vasudevan, Alexander Ku, Yinfei Yang, Burcu Karagol Ayan, et al. “Scaling autoregressive models for content-rich text-to-image generation”. In: *Transactions on Machine Learning Research* (2022).

[3] Joshua Peterson, Krishan Aghi, Jordan Suchow, Alexander Ku, and Thomas Griffiths. “Sampling from object and scene representations using deep feature spaces”. In: *Journal of Vision* 18.10 (2018), pp. 403–403.

[4] Ryan Poplin, Pi-Chuan Chang, David Alexander, Scott Schwartz, Thomas Colthurst, Alexander Ku, Dan Newburger, Jojo Dijamco, Nam Nguyen, Pegah T Afshar, et al. “A universal SNP and small-indel variant caller using deep neural networks”. In: *Nature biotechnology* 36.10 (2018), pp. 983–987.

Peer-reviewed conference papers

[1] Alexander Y Ku, Thomas L Griffiths, and Stephanie CY Chan. “An Evolutionary Perspective on Modes of Learning in Transformers”. In: *The Fourteenth International Conference on Learning Representations*. 2026.

[2] Andrew K Lampinen, Arslan Chaudhry, Stephanie CY Chan, Cody Wild, Diane Wan, Alex Ku, Jörg Bornschein, Razvan Pascanu, Murray Shanahan, and James L McClelland. “On the generalization of language models from in-context learning and finetuning: a controlled study”. In: *Foundations of Reasoning in Language Models Workshop, NeurIPS*. 2025.

[3] Declan Campbell, Sunayana Rane, Tyler Giallanza, Nicolò De Sabbata, Kia Ghods, Amogh Joshi, Alexander Ku, Steven M Frankland, Thomas L Griffiths, Jonathan D Cohen, et al. “Understanding the Limits of Vision Language Models Through the Lens of the Binding Problem”. In: *The Thirty-Eighth Annual Conference on Neural Information Processing Systems*. 2024.

[4] Yasumasa Onoe, Sunayana Rane, Zachary Berger, Yonatan Bitton, Jaemin Cho, Roopal Garg, Alexander Ku, Zarana Parekh, Jordi Pont-Tuset, Garrett Tanzer, et al. “DOCCI: Descriptions of Connected and Contrasting Images”. In: *The 18th European Conference on Computer Vision ECCV 2024*. 2024.

[5] Sunayana Rane, Alexander Ku, Jason Baldridge, Ian Tenney, Tom Griffiths, and Been Kim. “Can Generative Multimodal Models Count to Ten?” In: *Proceedings of the Annual Meeting of the Cognitive Science Society*. Vol. 46. 2024.

[7] Siddhartha Datta, Alexander Ku, Deepak Ramachandran, and Peter Anderson. “Prompt expansion for adaptive text-to-image generation”. In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2023.

[8] Zi Wang*, Alexander Ku*, Jason Baldridge, Thomas L Griffiths, and Been Kim. “Gaussian Process Probes (GPP) for Uncertainty-Aware Probing”. In: *Thirty-seventh Conference on Neural Information Processing Systems*. 2023.

[9] Aishwarya Kamath, Peter Anderson, Su Wang, Jing Yu Koh, Alexander Ku, Austin Waters, Yinfei Yang, Jason Baldridge, and Zarana Parekh. “A New Path: Scaling Vision-and-Language Navigation with Synthetic Instructions and Imitation Learning”. In: *The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2023*. 2022.

[10] Alexander Ku*, Peter Anderson*, Jordi Pont-Tuset, and Jason Baldridge. “Pangea: The panoramic graph environment annotation toolkit”. In: *Proceedings of the Second Workshop on Advances in Language and Vision Research*. 2021.

[11] Jiahui Yu, Xin Li, Jing Yu Koh, Han Zhang, Ruoming Pang, James Qin, Alexander Ku, Yuanzhong Xu, Jason Baldridge, and Yonghui Wu. “Vector-quantized Image Modeling with Improved VQGAN”. In: *The Tenth International Conference on Learning Representations*. 2021.

[12] Ming Zhao, Peter Anderson, Vihan Jain, Su Wang, Alexander Ku, Jason Baldridge, and Eugene Ie. “On the evaluation of vision-and-language navigation instructions”. In: *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*. 2021.

[13] Alexander Ku*, Peter Anderson*, Roma Patel, Eugene Ie, and Jason Baldridge. “Room-across-room: Multilingual vision-and-language navigation with dense spatiotemporal grounding”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2020.

[14] Haoshuo Huang, Vihan Jain, Harsh Mehta, Alexander Ku, Gabriel Magalhaes, Jason Baldridge, and Eugene Ie. “Transferable representation learning in vision-and-language navigation”. In: *Proceedings of the IEEE/CVF international conference on computer vision*. 2019, pp. 7404–7413.

[15] Vihan Jain*, Gabriel Magalhaes*, Alex Ku*, Ashish Vaswani, Eugene Ie, and Jason Baldridge. “Stay on the Path: Instruction Fidelity in Vision-and-Language Navigation”. In: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. 2019.

[16] Gabriel Magalhaes, Vihan Jain, Alexander Ku, Eugene Ie, and Jason Baldridge. “General evaluation for instruction conditioned navigation using dynamic time warping”. In: *Advances in Neural Information Processing Systems Workshop on Visually Grounded Interaction and Language*. 2019.

[17] Niki Parmar, Ashish Vaswani, Jakob Uszkoreit, Łukasz Kaiser, Noam Shazeer, Alexander Ku, and Dustin Tran. “Image Transformer”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.

[18] Joshua C Peterson, Jordan W Suchow, Krisha Aghi, Alexander Y Ku, and Thomas L Griffiths. “Capturing human category representations by sampling in deep feature spaces”. In: *Proceedings of the 40th Annual Meeting of the Cognitive Science Society (CogSci 2018)*. 2018.

Preprints

[1] Elizabeth Mieczkowski, Alexander Ku, Tiwalayo Eisape, Dilip Arumugam, John Matters, Katherine M Collins, Ilia Sucholutsky, and Thomas L Griffiths. “Improving the Efficiency of Language Agent Teams with Adaptive Task Graphs”. In: *arXiv preprint arXiv:2605.06320* (2026).

Patents

[1] Deepak Ramachandran, Alexander Ku, Peter James Anderson, and Siddhartha Datta. *Generating images by expanding prompts using generative neural networks*. US Patent App. 18/950,898. May 2025.

[0] Jiahui Yu, Vijay Vasudevan, Alexander Yeong-shiuh Ku, Yonghui Wu, Jason Michael Baldridge, Yuanzhong Xu, Jing Yu Koh, Thang Minh Luong, Gunjan Baid, Zirui Wang, et al. *Vector-Quantized Image Modeling*. US Patent App. 18/698,997. Dec. 2024.

Mentorship

2025–	Nicholas Budny (Undergraduate, Princeton) *Cross-task contamination in meta-learning*
2025–2026	Shirly Yu (Undergraduate, Princeton) *Resource-rational forgetting in continual learning*
2025–2026	Ryder Walsh (Undergraduate, Princeton) *Episodic memory for temporal credit assignment*
2025	Declan Campbell (Student Researcher, Google DeepMind) *In-context learning for mitigating catastrophic forgetting*
2024	Raja Marjieh (Student Researcher, Google DeepMind) *Probing semantic representations in large language models*
2024	Siddhartha Datta (Research Intern, Google Research) *Prompt expansion for adaptive text-to-image generation*
2023–2024	Sunayana Rane (Research Intern, Google Research) *Probing numerosity in vision-language models*
2019	Roma Patel (Research Intern, Google Research) *Multilingual vision-language navigation*

Teaching

2025	TA, Computational Models of Cognition, Princeton
2024	TA, Cognitive Neuroscience, Princeton
2024	TA, Developmental Psychology, Princeton
2017	TA, Data Science, UC Berkeley
2017	TA, Artificial Intelligence, UC Berkeley
2016	TA, Data Science, UC Berkeley

Service

2025–2026	Psychology Graduate Student Committee, Princeton
2018–	Reviewer (NeurIPS, ICLR, ICML, ACL, EMNLP)

References

Advisor	Thomas L. Griffiths (Professor of Psychology & Computer Science, Princeton) tomg@princeton.edu
Advisor	Jonathan D. Cohen (Professor of Psychology & Neuroscience, Princeton) jdc@princeton.edu
Manager	Michael C. Mozer (Research Scientist, Google DeepMind) mcmozer@google.com
Manager	Jason M. Baldridge (Research Scientist, Google DeepMind) jridge@google.com